

Integrating Neural Networks into Resource-Constrained IoT Devices: Challenges and Techniques

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Abstract—Abstract

Index Terms—component, formatting, style, styling, inserts

I. INTRODUCTION

Traditional statistical machine learning models have been an integral part of IoT device infrastructure. These models are generally memory-efficient, computationally inexpensive, and capable of achieving fairly accurate predictions in many practical scenarios. Due to these characteristics, they have historically been preferred for deployment in resource-constrained IoT environments. Among these statistical models, Random Forest classifiers and similar decision forest methods have consistently performed well for a variety of IoT tasks. These models are particularly effective in scenarios where the available datasets are relatively small, as they tend to remain robust even when training data is limited [1], [2]. When the underlying data is simple, well-structured, and easily interpretable, such models are often more than sufficient for achieving reliable performance in IoT applications.

However, these traditional machine learning models often struggle to learn complex representations when the underlying data itself is highly complicated. Existing literature notes that although decision forests are memory-efficient and practical for constrained environments, they remain limited in expressiveness. As a result, they frequently require dynamic adaptations, such as pruning strategies or meta-learning mechanisms, in order to handle more complex patterns and concept drift in data streams [3]. Similar limitations can be observed in other statistical machine learning models that rely heavily on handcrafted features and relatively shallow model structures. Furthermore, when the input data becomes high-dimensional or unstructured—such as visual, auditory, or multimodal data—these statistical models often fail to scale effectively, which significantly restricts their ability to capture deeper patterns within the data [4].

This is precisely the domain in which neural networks excel. Neural networks (NNs) are capable of learning complex hierarchical representations directly from raw data, enabling them to model intricate relationships that traditional statistical models often fail to capture. In addition, neural networks demonstrate strong scalability with respect to both the number of model parameters and the size of the datasets used during training [5].

Their representational power is further enhanced through the use of activation functions that introduce non-linearity into the learning process. Moreover, various architectural innovations like Convolutional Neural Networks (CNNs) for visual data [6] and Transformer architectures for sequence modeling [7] have significantly expanded the ability of neural networks to learn, generalize, and perform effectively across a wide range of complex tasks. As a result, neural networks have become a dominant approach for processing high-dimensional and unstructured data generated by modern IoT systems.

However, despite their potential, there are certain challenges that are prominent with neural networks. Efficient NNs are usually large [8] and therefore cannot be efficiently integrated to lightweight, hardware-restricted IoT devices. IoT devices typically have limited memory, storage and battery life. They also lack powerful processors and GPUs. Furthermore, raw data transfer between cloud, routers and IoT devices can be expensive as well. This greatly discourages integration of NNs in IoT devices and reduces their practical utility. As a result, substantial research efforts have been made to effectively integrate NNs in lightweight IoT devices.

Numerous developments have been made for lightweight integration of deep learning in resource-constrained environments, targetting different aspects of the deep learning ecosystem like hardware, network architecture, deployments and more [9]. However, memory, performance and hardware constraints of IoT devices make the integration even more challenging and more harmonized and considerable research is required in this aspect. This survey attempts to fill that gap.

II. IOT LANDSCAPE AND CONSTRAINTS

A. Device Tiers

Based on the tasks and their capabilities, we are going to divide IoT devices into three tiers. Some of them are more capable than the others. Each have their own place in the overall IoT architecture.

1) *Tier 1: Ultra-constrained Microcontrollers:* These IoT devices have the smallest SRAM and flash memory. The SRAM typically ranges from 16 to 512kb while flash storage is typically less than 2MB. The power consumption is less than 1mW in active mode and have no operating systems. Device families like ARM Cortex M0/M0+, ultra low-power

NPU and RISC-V ultra-constrained MCUs come under this tier. The TinyML framework primarily targets devices under this tier [10].

2) *Tier 2: Capable Processors*: Devices like Raspberry Pi, Nvidia Jetson Nano, and ARM Cortex-A based SOCs can be categorized under this tier. The RAM ranges from hundred megabytes to a few gigabytes with optional hardware accelerators. These devices allow us to deploy and infer deep learning models that are more accurate and efficient. Such devices can run minimal operating systems like RIOT OS [11]. Models like MobileNet and ResNet can be used for inference on these devices.

3) *Tier 3: Edge Servers and Gateways*: Devices within this tier represent the most capable class of hardware, typically offering gigabytes of memory, substantial storage capacity, and highly efficient high-performance processors. These platforms are capable of running full-scale deep learning models, including convolutional neural networks (CNNs) and transformer-based architectures. Commonly deployed in edge AI systems and cloud-connected infrastructures, they often serve as aggregation points for data collected from lower-tier IoT devices. They are also able to run operating system like Linux. As these devices are already fairly capable of running heavy models, these devices are not a primary focus of this survey.

In a typical IoT architecture, all these devices intercommunicate with each other. Among the hardware tiers discussed, ultra-constrained microcontrollers present the greatest challenges for deep learning deployment because of their severe limitations in memory, storage, processing capability, and energy availability. As a result, much of the recent research in TinyML, embedded AI, and resource-aware model optimization has focused specifically on enabling intelligent inference on these highly constrained devices.

B. Key Constraints

Here are some key constraints faced by IoT devices:

1) *Memory*: SRAM is the dominant bottleneck. Ultra-constrained MCUs rely solely on SRAM for computation. Standard deep learning models generate activation tensors that can easily exceed SRAM capacity.

2) *Computation*: MCUs (Microcontroller Units) usually have a clock cycle of 48MHz to 480MHz and limited instruction parallelism. This forces our hand to carry out MAC (multiply-accumulate) more efficiently within limited constraints. Architectures like SqueezeNet [12] and MobileNet [19] can reduce the number of operations, making such compressed architectures more suitable for IoT devices.

3) *Energy*: Most IoT deployments utilize battery-powered or energy-harvesting devices that impose strict power budget. Energy consumption during inference and training depends on the memory, numerical precision and arithmetic computation. Furthermore, DRAM consumes way more energy per bit compared to SRAM.

4) *Numerical Precision*: Typically, models use 32-bit precision floating points but that consumes a lot of SRAM. To fit such models within microcontrollers, some kind of

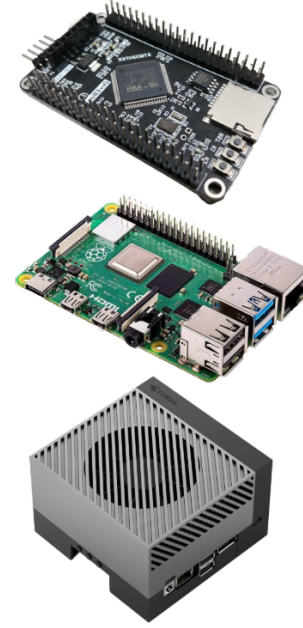


Fig. 1. Device of each tier. 1st one is STM32, 2nd Raspberry PI 4, and 3rd is Nvidia Jetson AGX Orin

compression is required. Quantization and pruning can help us out here, with a trade-off being the accuracy of those models.

5) *Storage*: The weights of the model reside in a flash memory, where a typical flash memory consists of 512kb to 8MB. Models like ResNet-50 [14] requires around 98MB of storage with float32 precision. Quantization to 4-bit compresses the size to 12MB which is still a lot for MCUs. Hence, pruning and low-bit quantization is an important requirement for compression.

6) *Latency*: Most of the IoT devices for anomaly, object and gesture detection require a response of milliseconds. Hence, the models deployed should meet the real-time requirements of these IoT devices, with quick inference and minimal latency.

III. IMPLEMENTATION STRATEGIES

Based on these constraints, the implementation, optimization, and deployment of deep learning models can be approached from multiple directions. The key areas of focus include neural network architectures, model compression strategies, hardware advancements and their suitability for AI integration, as well as lightweight frameworks that enable efficient deployment of AI on MCUs.

In this survey, we analyze these strategies in terms of their computational complexity and suitability for MCU deployment. We further examine the trade-offs between computation, memory usage, performance, and accuracy to identify practical operating points and highlight any standout approaches. Existing benchmarks are catalogued based on their computational requirements, performance, and accuracy trade-offs, with the analysis conducted primarily through the lens of IoT devices.

IV. NETWORK ARCHITECTURE

Our first area of interest in network architecture, where various neural networks of different compositions are introduced. Some architectures excel in accuracy while others chiefly focus on performance with minimal parameters. When it comes to vision tasks, CNNs are fairly competent on lightweight IoT devices, providing a favorable balance between performance and accuracy. However, since the introduction of transformer architecture [7], vision transformers have gathered a lot of attention and growing amount of research is being done to make them more lightweight and efficient. First, we outline models utilizing CNN architecture, their trade-offs and then the vision transformers.

A. Lightweight CNNs

1) *SqueezeNet*: SqueezeNet was introduced in 2016 with the goal of matching AlexNet's [15] performance with greatly reduced parameter count with minimal memory footprint [12]. SqueezeNet achieves this reduction by using 1×1 convolution filters instead of 3×3 convolutions. The squeeze layers reduce the number of input channels to subsequent filters. The resulting model has a memory footprint of approximately 4.8MB. This model can be further compressed with deep compression [16] (pruning, quantization to 6-bit and Huffman coding), resulting in a light 0.48MB model. This deep compression achieved comparable performance to the uncompressed one. While uncompressed SqueezeNet is not suitable for MCUs, there have been deployments of SqueezeNet-like CNNs on microcontrollers like Arduino Portenta H7 Lite [17], making SqueezeNet-inspired architectures a potential candidate for tier 1 devices.

2) *ShuffleNet*: ShuffleNet achieves reduction in model parameters and computation (FLOPs) by using group convolution and channel shuffling [18]. Instead of connecting every input channel to output channel, channels are split into groups which greatly minimizes computation and parameters. However, because of this grouping, each group sees its own channel which results in loss of information as there is no cross-mixing of the channels. To compensate, the channels are shuffled in the groups. Because of these operations, minimum accuracy is lost with reduction in computation and parameters. Because of this optimization, ShuffleNet is a strong candidate among IoT devices because of low FLOPs and storage requirement, especially the quantized and pruned versions.

3) *MobileNet*: The MobileNet series of models is a lightweight architecture of Neural Networks that achieved fairly high accuracy by reducing the computation at inference and lowering the number of parameters [19]. The core innovation of this architecture came from separation of spatial filtering and channel mixing using small filters, which resulted in high accuracy with reduced computation and size. Comparing in to SqueezeNet, the amount of FLOPs (Floating Point Operations) is greatly reduced in MobileNet which makes it even more suitable for lightweight devices with deep compression, as it has been deployed to MCUs like STM32H7 [21]. The latest release of MobileNetV4 further enhanced the

accuracy by utilizing Universal Inverted Bottleneck (UIB) and Mobile Multi-Query Attention (MQA) [20]

4) *AddressNet*: In the paper of AddressNet, two limitations are highlighted of other CNN architectures; MobileNet and ShuffleNet. It states that the shuffling of channels in ShuffleNet uses zero FLOPs but still consumes 24% of the inference time, and MobileNet's depthwise convolution takes up 20% of inference time, despite using less parameters [23]. To overcome this latency, AddressNet introduces pointer shifts instead of direct memory shifts. This greatly reduces the expensive memory access, decreasing the inference time. This pointer shifting can be done for channel shuffling. Furthermore, AddressNet replaces depthwise convolution by shifting the pointer of the feature map in four directions (up, down, left, right). While AddressNet is not used directly in MCUs, variants and different modifications over this architecture have shown great potential.

5) *EfficientNet*: Earlier, the CNN architecture focused on making it wider, deeper or using high resolution images for input. Instead of focusing on only one dimension, EfficientNet argues to scale along all the dimensions in a balanced manner [24]. This balance helps achieve reduced parameter size, better inference and higher accuracy. Similar to other architectures, vanilla EfficientNet is not suitable for MCU deployment but quantized and pruned versions are viable for microcontrollers with minimum accuracy loss.

6) *GhostNet*: The authors of GhostNet observed that many of the feature maps learned by CNNs are redundant. Instead of using standard convolution to compute those feature maps, they introduced 'ghost modules' by simple linear transformations of existing feature maps with minimal information loss. This transformation keeps the computation and number of parameters reduced while achieving competent accuracy to that of MobileNetV3 [25]. However, original GhostNet models are relatively large for microcontrollers. Pruning and quantization are a necessity to make them microcontroller feasible.

7) *MCUNet*: Instead of treating inference engine and neural network architecture as two separate tasks, MCUNet designs both the system as well as the model [26]. MCUNet introduces TinyNAS (Tiny Network Architecture Search) and TinyEngine. TinyNAS optimizes the search space itself based on the hardware - input resolutions, width multipliers, kernel sizes and other architecture options. TinyEngine is the inference engine. Unlike TensorFlow Lite Micro and other frameworks that have big overhead as they are interpreter-based, TinyEngine compiles the exact model instead of interpreting it at runtime. This reduces SRAM, flash overhead and latency. With these optimizations and quantization, MCUNet is easily deployable on microcontrollers. MCUNet was the first neural network on microcontroller to achieve more than 70% accuracy on ImageNet Top-1 classification. The latest release in MCUNet series is MCUNetV2 which further improves the memory consumption and other parameters of vanilla MCUNet, making it more lightweight as well as more accurate [27].

TABLE I
COMPARISON OF LIGHTWEIGHT CNN MODELS FOR MCU DEPLOYMENT

Model	Year	Parameters	Top-1 Acc. (ImageNet)	Memory (MB)	MCU Deployed?	Evidence
SqueezeNet [12]	2016	1.25M	57.5%	4.8	No	No MCU deployment found
SqueezeNet (deep compressed) [12]	2016	1.25M	57.5%	0.47	Yes	[17]
ShuffleNet v1 [18]	2017	~1.42M ^a	67.6%	~7.5	No	ARM mobile CPU only [18]
MobileNet v1 1.0 [19]	2017	4.2M	70.6%	~16.8	No	No MCU deployment found
AddressNet-20 [23]	2018	0.08M	68.68%	~0.32	No	No MCU deployment found
MobileNet v1 (mixed precision) [21]	2019	4.2M	68.0%	~2.0	Yes	STM32H743 [21]
EfficientNet-B0 [24]	2019	5.3M	76.3%	~21.2	No	Too large for MCUs
GhostNetV1 [25]	2020	5.2M	75.7%	~20.8	No	Too large for MCUs
MCUNet (4-bit) [26]	2020	~4M	70.7%	1.9	Yes	STM32 Family [26]
MCUNetV2 (8-bit) [27]	2021	-	71.8%	~2	Yes	STM32 Family [27]
MNV4-Conv-S [20]	2024	3.8M	73.8%	~15.2	No	Primarily for mobile CPUs/GPUs

^a Estimated from architecture table [22]; not reported in [18].

B. Transformers

V. COMPRESSION TECHNIQUES

A. Pruning

Pruning removes neurons, filters or even entire layers that are doing little for the neural network. The intuition being, neural networks are usually quite dense where most of the neurons are either redundant or doing barely anything for efficiency of the network. Pruning tries to remove those redundant weights, weak neurons and unimportant channels.

The earliest implementation of pruning involved calculating second order derivatives to approximate the unimportant neurons and then removing them from the network [28]. One limitation in this approach is that the weights are reduced at initialization, making the removal of weights a little less justified. However, post-training pruning has been quite successful [29]. In this technique, a neural network is first trained and then later on weights are removed without major sacrifices to accuracy. Then, the remaining network is re-trained for more robustness. These approaches highlighted come under the category of unstructured pruning, where individual weights are removed irrespective of their position in the tensor. This creates irregularities in connectivity and hardware improvements are limited.

On the other hand, structural pruning removes entire structural units of the neural network, like filters or channels in CNNs or fully connected layers in DNNs (Dense Neural Networks). Because of entire removal of structural units, the sanctity of the structure remains and we gain boost in hardware performance without any special sparse kernels. Early implementations of structural pruning involve removing filters that are less significant, based on the criteria of L1 norm [30].

Extensive analysis has been carried out to understand the impact of pruning on NNs. One hypothesis synthesizes that pruning can reduce the size of the network by 80 to 90% without sacrificing accuracy where specific modules have won 'the lottery ticket'. When a similar sized network is trained in isolation, the performance is worse. So, the lottery won by pruned model lies in the initialization of the neural network [31]. This hypothesis has directed the research for pruning, where finding 'the lottery ticket' has become a focus before after initialization and before training [32].

Pruning is one of the most prominent technique to make neural networks more microcontroller feasible. It has been used in conjunction with quantization and huffman coding to aggressively reduce model sizes [16]. Heavier model architectures like MobileNet and SqueezeNet have been pruned and deployed to MCUs [17].

B. Quantization

Quantization is the most prominent technique of greatly minimizing the model size by quantizing the data type of the neural network to less precision. For high accuracy, neural networks typically use single-precision 32-bit floating point (FP32) for weights. However, FP32 constitutes of 4 bytes. So, a model of 5M parameters is going to take up 20MB of storage, which is not viable for microcontrollers at all. Here, quantization comes into play, where the weights are quantized to a lower precision data type like INT8, INT4 etc, greatly reducing the model size.

The earlier impact of quantization can be seen in BinaryConnect, which showed that weights constrained to only -1 or +1 at extremely low precision can work for neural networks [33]. One of the influential studies and breakthrough in quantization was achieved by quantizing the floating points to integers [34]. Floating points have shown to be more compute intensive and slower at inference. But with integers, if done properly, quantization can achieve great reduction in model size without any major sacrifices to accuracy. The authors also introduced Quantization-Aware Training (QAT). Early implementation of quantization will take a model and quantize the learned weights later, which results in greater accuracy drops. With QAT, the model 'simulates' the impact of quantization in the forward pass. This makes models like MobileNet and EfficientNet more microcontroller friendly [16], [17], [26], [27], [34]. Integer-only quantization has become mainstream for TinyML and frameworks like TensorFlow-Lite. Mixed-precision quantization is also used for even lesser drops in accuracy, where some layers can use high-precision quantized integers while others utilize low-precision weights [21].

Quantization is easily one of the most extensively used technique to lower the memory footprint of neural networks,

increase hardware access and make inference faster, with minimal drops in accuracy.

C. Knowledge Distillation

In Knowledge Distillation (KD), a larger model (called teacher), trains a smaller model (students) uses its own internal representations. This is one of the most used technique to train high accuracy, small lightweight IoT devices. KD was introduced in 2014, with the idea that teacher model learns the hard target labels and reasons about it in soft probabilities. The student model then not only learns the 'softer' representations of the teacher, but also the hard labels, making it more concise and rich in learning [35]. This often achieves better convergence, generalization and robustness. While the student learns the output probabilities from the teacher, other approaches learn different parameters from the teachers. For example, FitNets extend the idea of classic KD where the student model learns not only the output probabilities but also the intermediary representations of the teacher [36].

Since then, there have been new variations of distillations introduced. In online distillation, teacher and student models are trained in conjunction, compared to classic distillation where teacher is trained first. In self-distillation, the teacher trains itself by using its own soft targets. The soft targets act as smoothening representations that improve the accuracy of the model, making it more 'intelligent' [37].

D. Network Architecture Search (NAS)

VI. FRAMEWORKS

A. CMSIS-NN

B. TensorFlow Lite Macro

C. TinyEngine

VII. APPLICATIONS

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